NOTES

TUT - 10

1). Bracelet Problem

2n Black 2n White

Define

For each $i = 1, 2, \ldots 4n$

let $Ar(i)$ be contiguous of 2n beads

Starting at i & going ↻

ending at $i + 2n - 1$, mod 4n.

$B(i) \rightarrow$ Black beads

$W(i) \rightarrow$ White $\rightarrow 2n - B(i)$

$B(i) - 1$

$i = \text{Black}$

$i + 2n \rightarrow$

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$i \rightarrow$ White

$i + 2n \rightarrow$ Black

$B(i+1) = B(i) + 1)$

$B(i)$

same colour for i & $i+2n$

$B(i+1) - B(i) \in \{-1, 0, +1\}$

$B(1) + B(2n+1) =$ total black beads $= 2n$

$B(2n+1) = 2n - B(1)$

$B(i) = n^*$

i = Black

$i+2n \rightarrow$

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$i \rightarrow$ White

$i+2n \rightarrow$ Black

same color

for i & $i+2n$

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Algo

Define $B(i)$ = black beads in arc of $2n$ starting at i

$B(1) + B(2n+1) = 2n$

$|B(i+1) - B(i)| \leq 1$

If $B(1) = n \rightarrow$ done

If $B(1) \neq n \rightarrow B(1)$ & $B(2n+1)$ lie on opposite sides of n.

Discrete IVT $\rightarrow \exists\, i^*$ so that $B(i^*) = n$

Cut at i^* & $i^* + 2n \rightarrow$ Both have exact n blocks of each

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2)

→ Recurence

$f(1) = 1$ base

$f(n) = \Sigma f(k) . f(n-k)$ for $n \geq 2$

$k = 1$ to $n-1$

Expand

$f(n) = f(1) . f(n-1) +$

$f(2) f(n-2) + \ldots$

$f(n-1) (f(1))$

Closed form is

$$f(n) = \frac{1}{n} . \binom{2(n-1)}{n-1}$$

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Base

$f(1) = 1$

$f(2) = f(1) * f(1) = 2$

$f(3) = f(1) . f(2) + f(2) f(1)$

$= 5.$

Cantlan formula

$$C_n = \frac{1}{n+1} . \binom{2n}{n}.$$

$f(1) = 1$

$f(n) = \Sigma f(k) . f(n-k)$

$$\boxed{f(7) = 686}.$$